

3(a) The work done by a force on an object particle is defined as the product of the magnitude of the force and the component of the displacement in the direction of the force. B1

3(b)(i) Change in KE = Net work done

$$0 - \frac{1}{2}mu^2 = -\frac{1}{2}F_{\max}x$$

M1

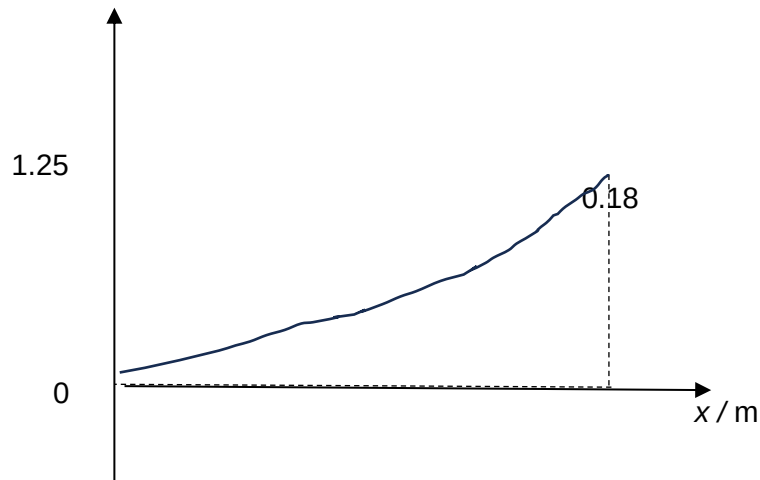
$$(0.400)(2.5)^2 = (14)x$$

$$x = 0.18 \text{ m}$$

A1

(ii)

work done by F / J



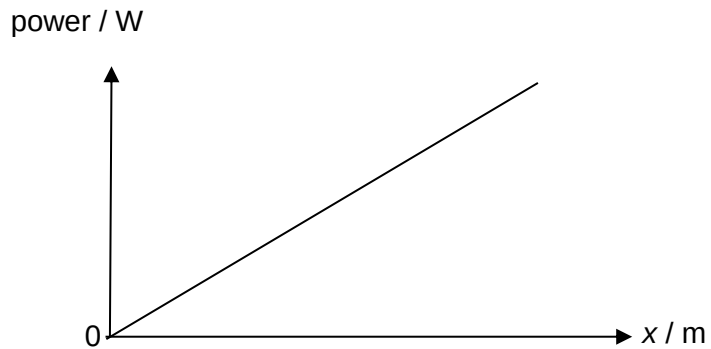
B1

B1

B1 – shape

B1 – x and work done values

(iii)



B1

Since F proportional to x , and v is constant, power also proportional to x

4 (a)

The direction of the object changes continuously and hence there is a change M1